Research records and data management plan

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STEM research placement · 16 September – 28 October 2026

Placement deliverable: data / lab book / experimental records

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Scope of this document

This document is the experimental record for the placement reported in *Buying back communication with retrieval compute in federated fine-tuning*. It describes what was collected, when and how; how the data are organised, verified and archived; what is kept and what is deliberately not kept; and where the reasoning behind each decision is preserved. It is written so that a reader who has never seen the project can locate any number in the report and regenerate it.

Placement period: **16 September 2026 – 28 October 2026** (six weeks). All work was performed by the author unless stated otherwise.

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1. Overview of the record

Item	Value
Experimental runs	24 (6 reranking depths × 4 seeds)
Main sweep executed	12 October 2026 – 20 October 2026
Wall-clock compute	33.9 h total, 1.41 h per run (mean)
Per-round logs	24 JSONL files, 24 JSON summaries, 868 KB
Pilot-gate artefacts	6 JSON files
Frozen input artefacts	7 JSON files (corpus, partitions, indexes, retrieval caches)
Analysis artefacts	1 JSON (`exchange_rate.json`, 300 KB)
Figures	4 PNG
Source modules	37 Python modules under `src/`
Analysis and utility scripts	29 under `scripts/`
Automated tests	333, all passing
Commits on the final branch	51

1.1 Nothing in this record is simulated or imputed

Every figure and table in the report is read directly from the archived run logs by a script in the archive. No value was transcribed by hand into the report without a script that reproduces it, and no missing value was imputed. A condition that failed to reach a target threshold is recorded as not reaching it, never extrapolated.

2. What was collected

2.1 Per-round experiment logs — the primary record

`results/logs/runs/d{depth}_s{seed}.jsonl`, one line per event.

The first line is a **manifest** fixing the conditions of that run: depth, seed, partition seed, rounds, local steps per round, total local steps, client count, batch size, gradient accumulation, examples per step, the evaluation round schedule, the early-stopping flag, the backend, the base model, trainable parameter count, payload bytes per client per direction, and the total communication budget with the formula that produced it.

Subsequent lines are **round records**: round index, whether the round was evaluated, held-out F1 and EM, evaluation-set size, gold recall at 3, cumulative bytes, wall-clock seconds, and mean client-update divergence.

`results/logs/runs/d{depth}_s{seed}_summary.json` holds the manifest plus final scores and treatment diagnostics (promotion rate, rank-change rate).

Logs are written fresh, not appended. An earlier version appended on retry, which produced curves that walked backwards in round order; the writer now truncates and the curve loader additionally de-duplicates by round, last write winning. The defect and its fix are preserved in the commit history.

2.2 Frozen input artefacts

Computed once and reused by every run, so that depth selects a different slice of identical inputs:

File	Contents
`results/logs/week2/corpus.json`	200,000-passage working corpus, build statistics
`results/logs/week2/partition_{1001,1002}_summary.json`	client allocation statistics for both partition seeds
`results/logs/week2/indexes_{1001,1002}.json`	dense index build records
`results/logs/week2/retrieval_cache_{1001,1002}.json`	frozen dense retrieval and cross-encoder scores

Freezing the retrieval cache is what makes the comparison clean: the cross-encoder scored each question once per depth rather than once per round, so no run differs from another by retrieval stochasticity.

2.3 Pilot-gate artefacts

`results/logs/gates/`:

File	Decision recorded
`gate1_headroom.json`	headroom 18.40 F1 (gold 19.57 vs random 1.17) ≥ 5.0 → proceed
`gate2_monotonicity.json`	gold recall at 3 rose 0.467 → 0.5095; recoverable share 0.202–0.307 ≥ 0.10 → proceed
`gate2_monotonicity_n300_gold2_superseded.json`	superseded 300-question version, retained
`gate3_local_steps.json`	divergence by K ∈ {4, 8, 16, 32, 64}; probe at 250 questions per client
`calibrated_schedule.json`	K = 32, S = 1,892, R = 59 re-derived at full scale
`scan_stats.json`	corpus scan: 84,076 of 91,535 questions covered

The superseded gate-2 file is kept on purpose. The first version used 300 questions and a weaker gold-matching rule; replacing it was a methodological decision, and deleting the earlier evidence would hide that the decision was made.

2.4 Derived artefacts

`results/logs/week4/flops.json` records the measured FLOPs per cross-encoder pair (5.437 × 10■, 256 tokens) and per embedded passage (5.437 × 10■, 128 tokens), each alongside its analytic estimate and the measured/analytic ratio, plus measured CPU throughput of 20.34 pairs per second. Recording both the measurement and the estimate means a reader can see that the cost model's denominator is not an unchecked assumption.

`results/logs/week4/main_runs.json` and `run_checks.json` record the sweep execution and the post-hoc integrity verdict. `results/logs/analysis/exchange_rate.json` is the single analysis artefact from which every table and figure in the report is produced.

2.5 Deliberately not retained

Per-round model checkpoints. They are large, fully derived from the logged configuration and the frozen inputs, and reproducible by re-running. Retaining them would multiply the archive size without adding evidence.

The base model and source datasets. `SmolLM2-135M-Instruct`, `nq_open` and `wiki_dpr_e5` are fetched from the Hugging Face Hub at run time and are not redistributed. Their identifiers and the exact column projection used are recorded in `configs/experiment_config.yaml`.

Intermediate embedding tensors. Regenerable from the corpus and the recorded embedding model.

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3. Provenance of inputs

Input	Source	Licence
Base model	`HuggingFaceTB/SmolLM2-135M-Instruct`	Apache-2.0
Questions	`google-research-datasets/nq_open`	CC BY-SA 3.0
Passages	`kenhktsui/wiki_dpr_e5` (mirror of DPR `psgs_w100`)	CC BY-SA 3.0 (Wikipedia)
Reranker	`cross-encoder/ms-marco-MiniLM-L-6-v2`	Apache-2.0
Embedder	`BAAI/bge-small-en-v1.5`	MIT

The CC BY-SA 3.0 provenance of the question and passage sources is the reason experiment logs are released under CC BY-SA 4.0 rather than Apache-2.0: generated text in the logs reproduces material derived from those corpora.

Ethics and permits. No human participants, animals, personal data or physical specimens were involved. The corpora are public research datasets used under their stated licences. No ethics approval was required, and none was sought.

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4. Inclusion and exclusion criteria

Recording these explicitly matters, because each one would otherwise be an invisible degree of freedom.

Gold passages. Included only when a normalised answer string was contained in the passage *and* the passage shared at least two discriminative content words with the question. Containment alone marks 98 % of the corpus as gold for some question, because 69 % of Natural Questions answers are a single token; the overlap requirement removed 96.8 % of those false positives.

Hard negatives. Up to five BM25 negatives per question, with answer-bearing candidates excluded so a distractor could not be accidentally correct.

Target thresholds. Of 80 candidate target values drawn from the baseline curves' own evaluated scores, 3 fell below the derived lower bound (18.13, the maximum over seeds of the baseline's round-one score) and 14 above the derived upper bound (28.14, the minimum over seeds of the baseline's peak), leaving 63. Both bounds are derived from the data, not declared.

Seeds. Four, not five. Two partition assignments are alternated across seeds, so only an even seed count balances the allocation; five seeds would give a 3:2 imbalance.

Runs excluded from analysis. None. All 24 completed and all 24 were used.

Outlier handling. None. The high-target tail of the saving curve swings by more than ±150 MB (Figure 2 of the report) and was **not** trimmed, because choosing a cut point would be a further arbitrary decision. Extreme-insensitive statistics — medians and counts of targets — were used

instead.

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5. Derived measures and transformations

Measure	Definition
Promotion rate	share of the three prompt slots holding a passage dense retrieval alone would not have selected
Rank-change rate	share of slots holding a different passage than the baseline, reordering included
Rounds-to-target	first round reaching target F1, by linear interpolation between evaluated rounds, computed **per seed then averaged**
Rounds saved	baseline rounds minus condition rounds, paired by seed
Bytes saved	rounds saved × 29.49 MB per round
Exchange rate	bytes saved ÷ reranking FLOPs
Break-even bandwidth	`saved × 8 × throughput ÷ FLOPs`, clamped to 0 when the numerator is non-positive
Client-update divergence	mean pairwise divergence of the eight client updates, averaged over the first ten rounds

Two ordering choices in this table are load-bearing and were both corrected during the placement after review. Interpolating per seed and then averaging — rather than averaging the curves first — preserves the between-seed variance the interval depends on. Taking the median of rounds saved across all 63 targets — rather than reading one representative target — prevents a single threshold from determining the result; an earlier draft read the exchange-rate numerator at one target, which gave two depths the opposite sign to the rounds table computed over all 63.

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6. Statistical decisions, and the ones that were revised

Pairing. Depth is compared with baseline seed by seed. Seed identity shifts a whole run's round count, and pairing cancels that component: 0.16–0.47 rounds paired against 0.879 unpaired.

Critical value. Four seeds give three degrees of freedom and a two-tailed 95 % critical value of **t = 3.182**. An earlier version of the analysis used one standard error in code and "twice the standard error" in prose — two different rules, both too lenient. Under the corrected rule the count of targets at which a saving is resolved falls from 26/10/16/9/7 to **5/0/1/1/5**, and one contrast that had read as significant (`d` = 10 vs `d` = 0) does not ($p = 0.127$). The direction of the conclusion was unchanged; the rule was wrong and is now right.

Resolution ceiling. A paired permutation test over four seeds has $2^4 = 16$ sign arrangements and therefore a smallest attainable two-tailed p of 0.125. This is recorded in the analysis artefact under `resolution_limit`, so that the `within_noise` verdict is read as a property of the design rather than as a failure to find something.

Verdict rule. A saving counts as observed only when it clears the decision threshold at a majority of the 63 targets. This rule replaced an earlier one under which any positive value counted; the earlier rule allowed a single +0.0988-round value with a standard error of 0.515 to flip the verdict from `no_effect` to `consistent`.

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7. How the record is organised

7.1 Weekly branches are the laboratory notebook

The repository carries linear branches `week1` … `week6`. Each preserves the state of the project at the end of that week, and commit messages carry the reasoning — what was tried, what failed, and why a choice was made — rather than only what changed.

Branch	Period	Content
`week1`	16 Sep 2026	plan, scaffolding, pilot gate 1
`week2`	Sep 2026	corpus construction, partitions, indexes, retrieval caches
`week3`	Sep–Oct 2026	gates 2 and 3, schedule calibration, pilot runs
`week4`	Oct 2026	main sweep (24 runs), FLOP measurement, integrity checks
`week5`	Oct 2026	analysis, cost model, mechanism check, report
`week6`	Oct 2026	review responses, corrections, final deliverables

This is the part of the record that cannot be reconstructed after the fact, and it is why branches are retained rather than squashed. Several decisions reported as single lines in the report — the gate-2 replacement, the interpolation ordering, the critical-value correction — are legible only in that history.

7.2 Repository layout

```
configs/     experiment configuration (single YAML, dev overlay)
src/         library code: data, retrieval, federated training, analysis
scripts/     runnable entry points and report conversion
tests/       333 automated tests
results/
  logs/
    gates/     pilot-gate artefacts
    week2/     frozen corpus, partitions, indexes, retrieval caches
    week3/     pilot run logs
    week4/     FLOP measurement, sweep record, integrity checks
    runs/      24 per-round JSONL logs + 24 summaries
    analysis/  exchange_rate.json
  figures/   4 PNG figures
docs/        weekly reports, final report, research records
```

7.3 Integrity verification

`MANIFEST.sha256` lists a SHA-256 for every archived file. To verify:

```
sha256sum -c MANIFEST.sha256
```

Separately, `scripts/check_runs.py` re-reads the 24 run manifests and asserts that rounds, client count, payload size, local steps, total bytes, batch size, gradient accumulation, examples per step and base model are identical across runs, and that the treatment diagnostic differs between conditions as it must. It reports **24/24 complete, identical budget, one axis varied**. A field recorded for some runs but not all is itself flagged, because a setting that was not recorded cannot be verified.

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8. Reproducing the analysis

Analysis runs from the archived logs alone and needs no GPU:

```
pip install -r requirements.txt
sha256sum -c MANIFEST.sha256
python scripts/check_runs.py
python scripts/analyze_exchange_rate.py
python scripts/plot_results.py
pytest tests/
```

`analyze_exchange_rate.py` rewrites `results/logs/analysis/exchange_rate.json`; `plot_results.py` rewrites the four figures from it. Every table in the report is derived from that artefact.

Reproducing the **runs** requires a GPU and roughly 34 hours:

```
python scripts/run_main_runs.py
```

A single run can be targeted with `--only-depth` and `--only-seed`. A changed base model requires `--set treatment.variant=<name>` so that its logs are written under distinct names; the runner refuses to proceed otherwise, because run names are built from depth and seed alone and would otherwise overwrite the main matrix.

Environment. Python 3.11 on Linux; training on an NVIDIA A10G GPU in bfloat16. The pinned package versions are in `requirements.txt` and a frozen environment listing accompanies the archive. Versions are pinned rather than ranged because `transformers` 5.x removed dataset script loading — which is why a parquet mirror of the passage corpus is used — and `sentence-transformers` 6.x renamed the embedding-dimension API; a ranged specification would silently run different code a few months later.

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9. Archiving and long-term storage

Citable snapshot. The repository is archived on Zenodo as a versioned release, under the same arrangement used for the author's previous placement (Zenodo record 22781474). The deposit is a single zip of the tagged snapshot, including `MANIFEST.sha256`, and is linked to the GitHub repository through an "is supplement to" relation.

Concept DOI (all versions)	**10.5281/zenodo.XXXXXXXX**
Version DOI (`v1.0.0`)	**10.5281/zenodo.YYYYYYYY**
Repository	https://github.com/hufs0529/rag_fl_STEM_placement
Release tag	`v1.0.0`

Zenodo mints two identifiers. The **concept DOI** resolves to whichever version is current and is the one cited in the report and in `CITATION.cff`; the **version DOI** pins the exact snapshot a given result was produced from. Both are recorded here so that a reader who needs the precise bytes behind a number can get them, rather than only the latest state of the project.

Working repository. GitHub, `hufs0529/rag_fl_STEM_placement`, with the week branches retained.

Retention. Zenodo provides long-term preservation with a persistent DOI; nothing in this project depends on GitHub remaining available. The archive is self-contained for analysis: the logs, the scripts that read them and the tests that constrain those scripts are all inside it.

Licensing. Source code, scripts and tests under Apache-2.0 (`LICENSE`). Experiment logs under `results/logs/` under CC BY-SA 4.0 (`LICENSE-DATA`), for the provenance reason in Section 3. Neither the base model nor the source corpora are redistributed.

Citation. `CITATION.cff` in the repository root carries the citation metadata; `.zenodo.json` carries the deposit metadata (title, creator, ORCID, affiliation, keywords, licence, related identifiers).

Analytical scripts. Archived with the data rather than separately, so that the artefact and the code that produced it cannot drift apart. The 333 tests are included for the same reason: they record what the analysis is supposed to do, and several encode specific defects that were found and fixed so that they cannot silently return.

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10. Deviations from the original plan

Planned	Actual	Reason
4 depths × 5 seeds = 20 runs	6 depths × 4 seeds = 24 runs	five seeds give a 3:2 partition imbalance; depths 7 and 25 were added on mentor feedback that the ladder was too coarse to show a trend
Target accuracy fixed at one value	63 targets swept	a single target lets the choice determine the result; bounds are derived from the baseline curves

Planned	Actual	Reason
Uniform grid of target values	the baseline curves' own evaluated values	on a uniform grid, 15 of 25 points fell inside the round 0→1 interval and the measured saving came out exactly zero
K = 4 from the pilot probe	K = 32	the probe ran at 250 questions per client; the decision was re-derived at full scale from round-one divergence
Gate 2 on 300 questions	re-run on 2,000 with a stricter gold rule	containment-only matching marked 98 % of the corpus as gold

An additional 8-run comparison at a larger model scale was scoped and the tooling for it was built and tested, but the runs were not executed: the estimated 28–40 hours of GPU time fell outside the placement budget. The capability is retained in the archive so the comparison can be run later without rework.

11. Known defects found and corrected during the placement

Recorded because a record that lists only successes is not a record.

Defect	Consequence had it not been found	Status
Logger appended on retry	curves walked backwards in round order	fixed; writer truncates, loader de-duplicates
Sweep launched twice concurrently	two processes writing the same logs	fixed; PID lock, verified to refuse a second launch
`--only-depth` accepted values absent from the matrix	4 runs executed silently where 8 were intended	fixed; the runner exits with the available values
Verdict rule accepted any positive saving	one +0.0988-round value (s.e. 0.515) flipped the verdict	fixed; majority-of-targets rule
Exchange rate read at one target	two depths carried the opposite sign to the rounds table	fixed; median over all 63 targets
Decision threshold used 1–2 × s.e.	one contrast read as significant that is not	fixed; t(0.975, 3) = 3.182
Mechanism figure used a dual axis	a 0.77 % change rendered as a collapse, contradicting the text	fixed; two panels, divergence axis anchored at zero
Manifest did not record the base model	a re-run under a different model would have overwritten the main matrix	fixed; recorded, checked, and guarded
Batch split not recorded in the manifest	identical-budget claim unverifiable	fixed; three fields added and compared

12. Use of generative AI

An AI coding assistant was used throughout, consistent with Macquarie University policy. It wrote and refactored analysis code, drafted and revised report text, and reviewed statistical reasoning; several of the defects in Section 11 were identified in that review. All design decisions and all interpretation were made and approved by the author.

No experimental data were generated, altered or synthesised by an AI system. Every number in the report and in this document is produced by an archived script reading an archived log, and the full development history — including the reasoning behind each change and each correction — is preserved in the repository's commit record and weekly branches.